2024 PU3 H2 Physics Prelim Paper 2 Suggested Answers

1	**(a)(i)**	The vertical acceleration of the ball is constant in magnitude (9.81 ms^{-2}) and directed downwards throughout the path. Hence as the ball moves from the ground to the highest point, vertical velocity decreases at a constant rate till vertical velocity reaches zero. As the ball moves from the highest point towards the ground, vertical velocity increases at constant rate.	B1 B1 B1
	(a)(ii)	The horizontal acceleration of the ball is zero throughout the path. Hence the horizontal velocity is constant in magnitude.	B1 B1
	(b)(i)	Considering the ball just after it leaves the ground to the point just before it hits the ground, $v_y = u_y + a_y t$ $15 \sin 20^o = (-15 \sin 20^o) + (9.81)t$ $t = \frac{30 \sin 20^o}{9.81}$ $s_x = u_x t = (15 \cos 20^o)\frac{30 \sin 20^o}{9.81}$ = 14.7 m	 C1 C1 A1
	(b)(ii)	Considering the ball just as it drops down the cliff to just before it hits the ground. $s_y = u_y t + \frac{1}{2} a_y t^2$ $70 = -15 \sin 20^o t + \frac{1}{2}(9.81)t^2$ t = - 3.29 (NA) s or 4.34 s $s_x = u_x t - 14.7 = (15 \cos 20^o)(4.34) - 14.7$ $= 46.4\ m$	 C1 C1 C1 A1

2	(a)	Gravitational field strength is defined as the force per unit mass acting on a	A1